GEOGRAPHY 328 - Web GIS

Instructional Team:

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- **Alex Kirchmeier**, Teaching Assistant, alexak24@uw.edu | Office Hour: Thursdays 12:00 to 2:00 PM, Smith 428
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Web-based Geographic Information Systems (Web GIS) combine web, mobile technology, and GIS. It's a growing field in academia and industry, expanding GIS capabilities from local servers to the cloud. This makes online maps and geospatial data more accessible to fields like geography, transportation, and oceanography. The integration of web and GIS has led to advancements like smart cities, location-based services, and pandemic dashboards. This course teaches students to manage web GIS projects, use the latest geospatial cloud technologies, and provides real-world case studies.

This course will explore using ChatGPT to assist in Web GIS development. ChatGPT can simplify development, enhance productivity, and reduce the learning curve with its natural language interface. It helps with error analysis, data analysis, and producing GeoJson data for story maps, making web GIS applications more engaging. ChatGPT's user recommendations can personalize and improve the user experience. However, ChatGPT is not flawless—it may produce errors or reflect biases, so students are expected to critically evaluate and verify its outputs.

All Web GIS applications in the course can be used on both Windows and Mac OSX, with all required software being open-source or free. The course includes lectures and lab exercises. Lectures cover Web GIS theories like web architecture, front-end coding, responsive design, and web-based spatial analyses. Labs provide hands-on practice in web programming and web GIS development.

This web page is the syllabus. The schedule might change during the quarter, so check it frequently. If you have any questions, feel free to contact the instructor.

Course Objectives

- Understand the basics of web architecture, GIS project management, geospatial data clients, servers, and web-based spatial analysis.
- Learn to build web-based GIS applications using open-source or proprietary frameworks.
- Gain hands-on experience in web server management, coding with HTML, CSS, JavaScript, and using geospatial cloud technologies like MapBox and OpenStreetMap.
- Evaluate the user experience and social implications of real-world web GIS applications, including issues like geo-privacy and data authenticity.
- Use ChatGPT to assist in the design and development of WebGIS applications.

Weekly Schedule

To get started, you need to Gear up the working environment. ***Over this quarter, you are expected to read all the assigned reading materials by the end of the Tuesday of each week and complete the lab assignment and quiz by the due day***.

Week 1: Intro to Web GIS

The lecture sessions in this week introduces you to the fundamentals of Web GIS. Over this quarter, you will frequently use GitHub for synchronizing course material and managing Web GIS project. So, in Lab 1, you will learn to manage a web-based project using GitHub.

- **Readings:**
    - Intro to Web GIS
    - Git Handbook Optional. Git is an example of a distributed version control system (DVCS) commonly used for open source and commercial software development. With Git, Developers can work anywhere and collaborate asynchronously from any time zone.
    - Mastering Markdown Optional. Markdown is a lightweight and easy-to-use syntax for styling all forms of writing on the GitHub platform. You control the display of the document; formatting words as bold or italic, adding images, and creating lists are just a few of the things we can do with Markdown.
    - To grasp the gist of the reading material, you can refer to the study questions of this week.
- Lecture Slides
- **Lab 1:** Project management using GitHub
- **Thinkpiece 1:** Compare ChatGPT and GitHub Copilot

Week 2: Web Fundamentals

Today, almost everyone has used the Internet. To develop Web applications by yourselves, you need to dive into some fundamental concepts. So, in this week, you will learn the basics of the web environment, its mechanics and standards. To maximize your learning outcome, we encourage you to read over the reading materials before the lectures. To test how well you are familiar with the materials and the lecture content, you are required to complete the Quiz 1 all by yourself before the due day.

- **Readings:**
    - Getting started with the web. This document introduces you to the practicalities of web development.
    - The web and web standards. This article provides some useful background on the Web — how it came about, what web standard technologies are, how they work together, why "web developer" is a great career to choose, and what kinds of best practices you'll learn about through the course.

    - Common questions on Web mechanics. This document covers questions relating to general knowledge of the web ecosystem and how it works.
    - Technical basics of Web GIS
    - To grasp the gist of the reading material, you can refer to the study questions of this week.
- Lecture Slides
- **Quiz 1:** Web Basics

Week 3: Front-end Coding: HTML and CSS

To build websites, you should know about HTML, CSS and JavaScript. HTML is the fundamental technology used to define the structure of a webpage, CSS is used to style the web page, and JavaScript takes the charge of the behaviors of the Web. We will spend two weeks to focus on these three primary coding languages of the Web. This week mainly introduces you to HTML and CSS. To evaluate your learning outcomes, we offered a quiz and a lab for your practice.

- **Readings:**
    - Introduction to HTML. This document sets the stage, getting you used to important concepts and syntax, looking at applying HTML to text, how to create hyperlinks, and how to use HTML to structure a webpage. *The section on "Debugging HTML" and the two following assessments are not required*.
    - Multimedia and embedding This document explores how to use HTML to include multimedia in your web pages, including the different ways that images can be included, and how to embed video, audio, and even entire other webpages. *The section on the assessment is not required*.
    - CSS First Steps CSS (Cascading Style Sheets) is used to style and lay out web pages — for example, to alter the font, color, size, and spacing of your content, split it into multiple columns, or add animations and other decorative features. This module provides a gentle beginning to your path towards CSS mastery with the basics of how it works, what the syntax looks like, and how you can start using it to add styling to HTML.
    - CSS Selectors In CSS, selectors are used to target the HTML elements on our web pages that we want to style. There are a wide variety of CSS selectors available, allowing for fine-grained precision when selecting elements to

style. In this article and its sub-articles we'll run through the different types in great detail, seeing how they work.

    - Introduction to CSS layout This article will recap some of the CSS layout features we've already touched upon in previous modules, such as different display values, as well as introduce some of the concepts we'll be covering throughout this module.
    - To grasp the gist of the reading material, you can refer to the study questions of this week.

- Lecture Slides
- **Thinkpiece 2:** Responsive Design with ChatGPT
- **Lab 2:** Responsive web page design

Week 4: Front-end Coding: Javascript and GeoJSON

In this week, we will focus on learning JavaScript. Also, for web applications, geographical data are stored in the JavaScript Object Notation (JSON) format, or namely GeoJSON. This week will introduce you to the general format of GeoJSON, demonstrate how to asynchronously load GeoJSON data to your Web GIS application. Similar to the previous week, we offer a quiz on JavaScript and a new lab on how to load, parse and map GeoJSON data on the web.

- **Readings:**
    - JavaScript First Steps. In this document,we first answer some fundamental questions such as "what is JavaScript?", "what does it look like?", and "what can it do?", before moving on to taking you through your first practical experience of writing JavaScript. After that, we discuss some key building blocks in detail, such as variables, strings, numbers and arrays.
    - Javascript Building Blocks. In this document, we continue our coverage of all JavaScript's key fundamental features, turning our attention to commonly-encountered types of code blocks such as conditional statements, loops, functions, and events. You've seen this stuff already in the course, but only in passing — here we'll discuss it all explicitly.
    - Working with JSON. JavaScript Object Notation (JSON) is a standard text-based format for representing structured data based on JavaScript object syntax. It is commonly used for transmitting data in web applications (e.g., sending some data from the server to the client, so it can be displayed on a

web page, or vice versa). You'll come across it quite often, so in this article we give you all you need to work with JSON using JavaScript, including parsing JSON so you can access data within it, and creating JSON.

    - Making asynchronous programming easier with async and await More recent additions to the JavaScript language are async functions and the await keyword, added in ECMAScript 2017. These features basically act as syntactic sugar on top of promises, making asynchronous code easier to write and to read afterwards. They make async code look more like old-school synchronous code, so they're well worth learning. This article gives you what you need to know.
    - GeoJSON is an open standard format designed for representing simple geographical features, along with their non-spatial attributes. It is based on the JSON format. In addition, a notable offspring of GeoJSON is TopoJSON, an extension of GeoJSON that encodes geospatial topology and that typically provides smaller file sizes.
    - To grasp the gist of the reading material, you can refer to the study questions of this week.

- Lecture Slides
- **Quiz 2:** Front-end Coding Basics
- **Lab 3:** Asynchronous GeoJSON data loading and visualization

Week 5: Geospatial Web Server

In this week, we will focus on geospatial web servers. A geospatial web server plays a significant role in maintaining web based geospatial application. It stays in the cloud and provides multiple services relevant to geospatial data, such as geospatial data indexing, data format conversation, reprojection, and even spatial analyses. In this week, two geospatial web servers are introduced, including Mapbox (Proprietary) and GeoServer (Open Source).

- **Readings:**
    - Web map service from GeoServer
    - Publish geospatial data on GeoServer
    - Here in a list of active learning activities that will be demonstrated in the lecture session.

- **Instructions for GeoServer Installation on Windows and Mac** If you are using Windows, please download the MSI file for the stable version of GeoServer and proceed with the installation. For Mac users, download the Platform Independent Binary version instead. Regardless of your operating system, it's essential to install Eclipse Adoptium's OpenJDK beforehand to ensure a successful GeoServer installation. Please ensure you download version 11 of OpenJDK, as the latest version (version 2*) is not compatible. Mac users can follow this tutorial for detailed instructions on installing GeoServer, as well as starting and shutting it down.

- **Thinkpiece 3:** Generating GeoJSON Data with ChatGPT
- **Final Project:** Introduce the requirements of the final project.

Week 6 : Thematic Map Design on the Web

In this week, we will introduce you to map making on the web. The web based interactive map is built upon MapBox, which is a popular map library. It not only contains fundamental map related functions, but also offers a lot of cool map features for your use and further explore. We will walk you through at least two MapBox applications, and you will use MapBox to make an interactive web map in the lab session too.

- **Readings:**
  - Making thematic map on the web
  - Add interactive makers to web maps list of active learning activities that will be demonstrated in the lecture session.
- **Lab 4:** Interactive web mapping
- **Final Project Checkpoint #1**: Project proposal

Week 7 : Geocoding on the Web

Starting from this week to the last week, we will dive into a series of lectures on web-based spatial analysis. In this week, we will introduce you to web-based geocoding using MapBox Geocoding API. The Mapbox Geocoding API allows you to make forward geocoding, which means that a text query like University of Washington gets turned into longitude and latitude coordinates. But sometimes it's not enough to find query results. Often, you want the geocoder to find query results that are biased toward a location, limited to a specific area, or both.

- **Readings:**

    - Local search with the Geocoding API

- **Final Project Checkpoint #2**: Data preparation

Week 8 : Sorting by Distance

In this week, we will introduce you to a basic spatial analysis that uses distance to sort geographical data. This application is made upon MapBox too. Starting from this week, you will work on your final project. Its requirement will be introduced in this week's lab session. In short, you will need to make an advanced spatial analysis to deal with a real-world problem.

- **Readings:**
    - Sort data by Distance: Part I
    - Sort data by Distance: Part II
    - Visualize geographical changes over time

Week 9 : More advanced spatial analysis on the web

In this week, we will introduce how to deal with time on the web. The instructor will walk you through a more complicated application that illustrate how geographical data changes over time. In the lab session, you will continue to work on your final project.

- **Readings:**
    - Nearest neighbor analysis
    - Buffer and isochrone analysis
- **Thinkpiece 4:** Spatial Analysis with ChatGPT

Week 10 : Project Development and Presentations

We will still meet on Tuesday for a project development session. I will be there to help you with any questions regarding your final project. What's more, we will hold our final project presentation on Friday during the lab session this week.

Final Project Instruction

Final Project Presentation: will be held on Friday, December 5th, during the lab sessions from 11:30am to 3:20pm PST.

Final Project Submission : needs to be delivered to Canvas by the next Monday, December 8th 11:59 PST.

In-Group Peer Evaluation: Each group member needs to submit the peer evaluation form on canvas by the next Monday, December 9th 11:59 PST.

Course Requirement

GitHub: This course material will be hosted on GitHub instead of UW Canvas. On this dedicated GitHub repository, you can find most of the course material, participate in group discussions by submitting GitHub issues, and create new GitHub repositories to turn in the lab deliverables. By the end of this quarter, you will be more proficient in operating a cloud-based coding environment and able to host your work online as a way to gain public and peer attentions.

Participation: You should participate in the ad-hoc discussion items and complete all assigned readings and get familiar with the lab instructions before class meetings.

Quizzes: There will be two quizzes in this quarter. Each quiz is designed to test your comprehension of new material. Questions may include multiple choice questions, matching questions, fill-in-the-blank questions, and short answer questions. Before you answer the questions, please read the following instructions carefully:

- **Please do not use ChatGPT or any other AI tools to answer the questions.**
- You'll have unlimited time.
- You are welcome to use your notes, course material, and online resources, but you are asked to work alone (not in consultation with your classmates).
- You will be given only one attempt at each quiz.
- After completing the quiz, the first time, you'll see which questions you got right and wrong and get feedback on your answer selections.

Lab Assignments: You need to finish all four labs by the due date. You are encouraged to work on the lab assignments in groups, However, each student should submit their own work. In addition, you are encouraged to use ChatGPT to code or debug your program, but please do not use ChatGPT to create textual content that is required in your Lab Assignments. If you have any questions about the lab further, please refer to the troubleshooting section below.

Thinkpieces: Your weekly thinkpiece should be at least 350 words and share it on the Canvas discussion board. Your thinkpiece should delve into your utilization of AI tools such as ChatGPT, GitHub Copilot, or any other relevant tools in completing the lab assignment.

- While you are encouraged to use ChatGPT for coding or debugging your program, do not use ChatGPT or any other AI writing-assistant tools to draft your thinkpiece. This piece should be completed by yourself.
- Once you have posted your thinkpiece, engage with your classmates' contributions by commenting on or liking at least two of their thinkpieces. To grade your work, I will divide the students into three groups based on the number of likes and comments they receive by the end of the following week. For instance, the thinkpiece submitted in the first week will be evaluated at the conclusion of the second week. Self-generated likes and comments will not be considered in the assessment.
- Students who receive the highest number of likes and comments will be awarded 7 points, while those who secure the second-highest number will earn 6 points. The remaining students who submitted a thinkpiece will receive 5 points.

Grading

Grading items	**%**
Participation	2%
Thinkpiece	18%
Quizzes	15%
Lab Assignments	40%
Final Project	25%

Troubleshooting

General Course Issues

For course-related issues, please first check the syllabus. If you still have questions, email me directly.

Coding Issues

If your issue is related to coding and debugging, please follow these steps before reaching out:

1. Ensure you have carefully read the course requirements.

2. Verify that your code follows basic syntax and rules.
3. Use the inspector or your development environment to check for system errors or warnings.
4. Search for solutions on Google or other search engines. If you find any, try implementing them.
5. If the issue persists, use ChatGPT to seek a solution.
6. If you have tried all the above steps and still cannot resolve the problem, email the TA. In your email, describe the issue and include:
    - The source code or a link to your GitHub repo.
    - System errors or warnings from step 3.
    - Links to solutions you found through Google (step 4).
    - Screenshots of ChatGPT's responses (step 5).
    - A subject line summarizing your email (e.g., "Question regarding Web Map Stylesheet of Lab 4").

Equity & Inclusivity

Our very highest priorities include creating a brave and supportive class environment where each of us contributes, we can ask big questions, we give and receive critiques in a supportive way, we notice and engage the ways that we are differently situated within past and present relationship of power, privilege and oppression. I invite you to think hard about how race, gender identity, religion, age, citizenship status, first language, ability, sexuality, class, and other axes are at work in our interactions, and what this might mean in terms of when to speak up, when to step back, how to listen, and much more. Each of you is a welcome and invaluable part of our collective whole.

Disability Accommodations

We welcome the opportunity to work with any students with disabilities in this class to ensure equal access to the course. If you have a letter from Disability Resources for Students (DRS) outlining your academic accommodations, please present the letter to me (or email us, to confirm, if the letter is electronic) as soon as possible so that we can discuss the accommodations you may need for this class. Any discussions between student and professor need to occur as early as possible in order for adequate arrangements to be made. If you do not yet have a letter from DRS, but would like to request academic accommodations due to a disability, please contact DRS here (Links to

an external site.), or in-person at 011 Mary Gates Hall, or at 206-543-8924 (Voice & Relay), mailto:uwdrs@uw.edu.

Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Religious Accommodations Policy. Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form.

Student Care & Safety

It is important that you take care of yourselves inside and outside of class as you work through stress and other obstacles. There are many different support services on campus that can help, such as the Counseling Center, Hall Health, and the IMA. UW's Student Care program can help you connect to these and other resources. Learn more an contact them directly: http://depts.washington.edu/livewell/student-care/, livewell@uw.edu, or 206.543.6085. If you are concerned about yourself or a friend who is struggling SafeCampus is a helpful resource. Please add 206.685.7233 to your phones

Copyright

This course advocates for the open culture. The course materials are open source for both students and open source community to access. This course also builds on the work of Bo Zhao, Xiaoqi "Steven" Bao, Liz Peng, among many others, who have designed and taught or TA-ed previous iterations of this course.

Notably, students are not allow to videotape or audio-tape (record) this class in any form, and sharing recordings outside of class without the written consent of each student in the class is not permitted by FERPA. I encourage each of you attend the lectures instead of watching the recorded videos afterwards. Your in-class participation is a key factor to yield the best learning outcome. The instructor determines if their class can and cannot be recorded. This decision should be clearly communicated by the instructor at the beginning and throughout the quarter. In Zoom, the recording feature can be controlled by the instructor, as the meeting host.